Classical ML Pipeline — Kidney CT Classification Report

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MODEL CONFIGURATION

```
Classifier   : SVM (scikit-learn SVC)
Kernel       : RBF   C = 1000.0   gamma = 0.001
Features     : 138 dimensions across 7 groups
               (1) intensity stats  (2) GLCM/Haralick  (3) multi-scale LBP
               (4) Gabor filter bank  (5) intensity histogram
               (6) pre-CLAHE raw intensity  (7) morphological bright-region
Augmentation : BorderlineSMOTE  (strategy=not majority, kind=borderline-1)
Class weight : balanced  |  Scaler: StandardScaler  |  PCA: disabled
```

DATASET

```
Dataset      : CT-KIDNEY-DATASET-Normal-Cyst-Tumor-Stone  (11,929 images)
Classes      : Cyst / Normal / Stone / Tumor
Split        : Train 8,146  |  Val 1,895  |  Test 1,888
Strategy     : Per-class group-based (group_size=50) — patient slices kept together
```

TEST-SET PERFORMANCE (held-out, n = 1 888)

```
Macro F1     : 0.9091   (95 % CI  0.8944 – 0.9229)
Accuracy     : 0.9280
ROC-AUC (OvR): 0.9674
Per-class F1 : Cyst 0.9612  |  Normal 0.9392  |  Stone 0.7991  |  Tumor 0.9367
```

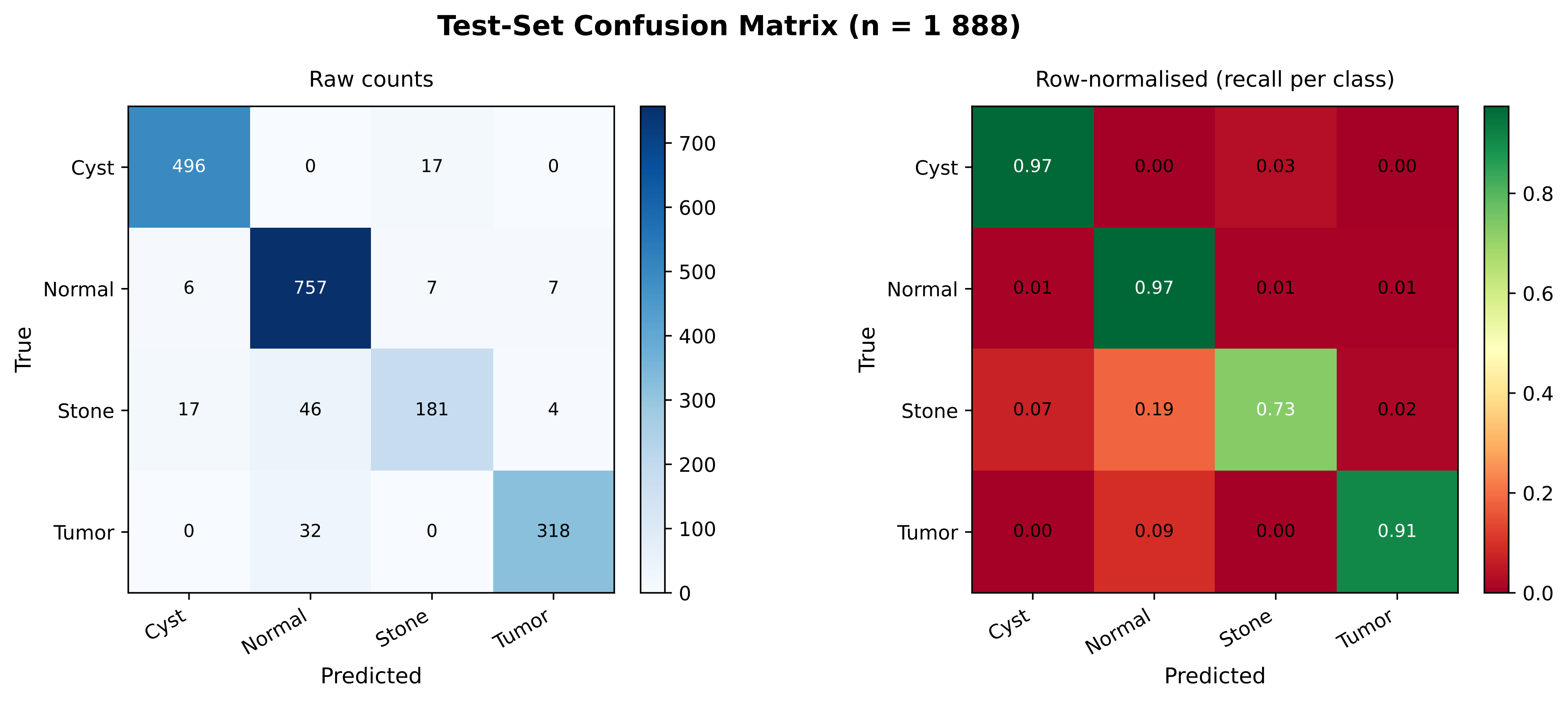
MODEL SELECTION (grid search)

```
5-fold StratifiedGroupKFold  |  Scoring: macro-F1  |  Candidates: SVM, RF
SVM best CV F1 : 0.8396   val F1 : 0.8015  ← WINNER
RF  best CV F1 : 0.8027   val F1 : 0.7943
Training time  : 37.3 min total
```

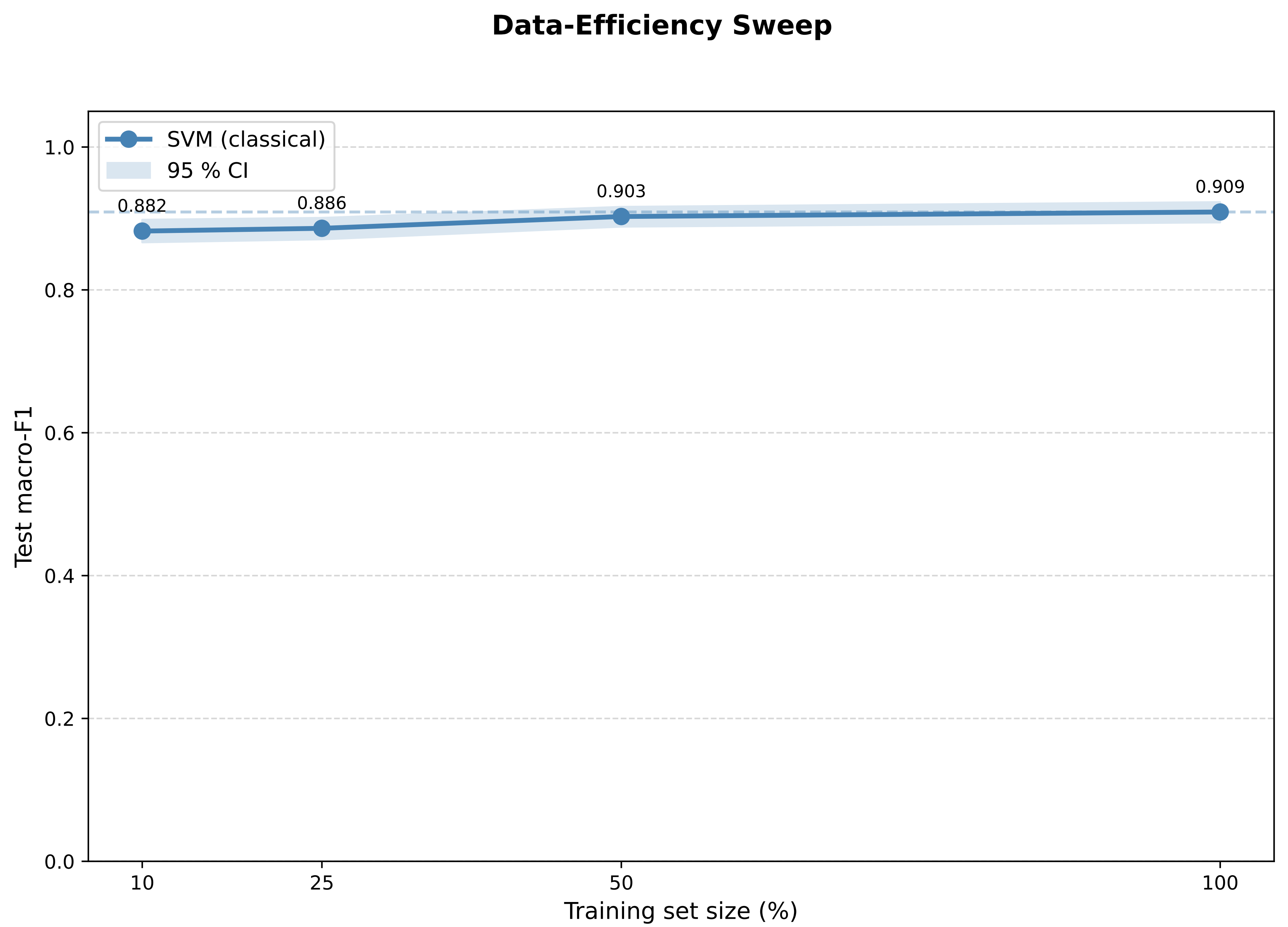
Per-Class Metrics — Train / CV (5-fold) / Test

Class	Train P	CV P	Test P	Train R	CV R	Test R	Train F1	CV F1	Test F1
Cyst	1.000	0.842	0.956	1.000	0.869	0.967	1.000	0.855	**0.961**
Normal	1.000	0.929	0.907	1.000	0.925	0.974	1.000	0.926	**0.939**
Stone	1.000	0.730	0.883	1.000	0.751	0.730	1.000	0.733	**0.799**
Tumor	1.000	0.884	0.967	1.000	0.826	0.909	1.000	0.845	**0.937**
Macro avg	**1.000**	**0.846**	**0.928**	**1.000**	**0.843**	**0.895**	**1.000**	**0.840**	**0.909**

Note: Train P/R/F1 = 1.000 for all classes. SVM with C=1000 fits training data exactly (low regularisation). High test F1 (0.909) confirms genuine generalisation — the model is not merely memorising.



Test-Set Confusion Matrix (n = 1 888)
Raw counts
Row-normalised (recall per class)
True
Predicted
Cyst
Normal
Stone
Tumor
496
0
17
0
6
757
7
7
17
46
181
4
0
32
0
318
0.97
0.00
0.03
0.00
0.01
0.97
0.01
0.01
0.07
0.19
0.73
0.02
0.00
0.09
0.00
0.91



Monotonically increasing F1 with training data indicates genuine learning, not memorisation.

Split Integrity & Leakage Audit

CHECK 1 — Filename overlap

Verifies that no image file path appears in more than one split.

```
train ∩ val  :  0 duplicates
train ∩ test :  0 duplicates
val   ∩ test :  0 duplicates
```

RESULT: PASS ✓

CHECK 2 — Exact feature-vector duplicates (bit-for-bit)

Extracts all 138 features from every image and hashes each row.

Any two images with an identical feature vector are flagged — catches

renamed or resaved copies of the same scan.

```
train ∩ val  :  0 exact feature matches
train ∩ test :  0 exact feature matches
val   ∩ test :  0 exact feature matches
```

RESULT: PASS ✓

CHECK 3 — Mean-intensity collision proxy

Images in different splits may share the same mean pixel intensity

(coarse bucket, rounded to 2 d.p.) — this is expected and benign.

train ∩ test shared buckets : 1 036 (out of 1 556 unique in test)

Interpretation: ~67% of test images share a mean-intensity bucket with

a training image — normal for CT data from the same anatomy.

Check 2 confirms none of these are actual duplicates.

RESULT: EXPECTED / BENIGN ✓

OVERALL: NO DATA LEAKAGE DETECTED